

第三届“认证杯”数学中国

数学建模国际赛

承 诺 书

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Research on the cost and control of the epidemic

Abstract:

In this paper, we mainly study the infection rate of Ebola in Uganda, estimate the deadline of producing new drugs and provide a reward policy for government to the pharmaceutical industries, which provides the project of combining pharmaceutical companies to improve efficiency and reduce risk. These studies raise the efficiency of government to deal with epidemics and have a positive impact on the maintenance of public health and safety.

After analyzing the Ebola infection cases in Uganda, we build a infection rate prediction model. We estimate the infection rate of Ebola from 2006 to 2025. Taking discount into consideration, the costs of resolving the crisis in 2013 should be 2076.47 million dollars.

Then, by analyzing the comprehensive index of other infectious diseases in this area during certain period, we build 'necessity of drug development' model to determine the necessity of new drugs. We analysis the infection rate and believe the research of new drug must be completed by 2008. And we estimate output of new drug with the method of analysis of variance at the same time.

In addition, based on government's investment, with the use of nonlinear programming method, we provide government a reward policy for pharmaceutical companies, including \$210million's government budget and \$207.3432million's public health services budget, to ensure the pharmaceutical process and sales.

Finally, we establish the indexes of pharmaceutical costs and risk, give the company a combination model, so that the government could utilize the index to improving efficiency of the pharmaceutical.

Key words: Ebola Budget GPM Nonlinear programming MATLAB

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1 Introduction

1.1 Background

A major outbreak of the Ebola virus across the African continent in 2014 triggers the global attention due to its huge number of infections and high mortality rates.

1.1.1 Ebola virus

Ebola is also called the Ebola virus. It is a very rare virus, being discovered in southern Sudan and the Congo (DRC) of the Ebola River in 1976. This is a potent infectious virus which can cause humans and primates produce hemorrhagic fevers. Ebola has a high mortality rate between 50-90%. The main cause of death was a stroke, myocardial infarction, low hemorrhagic shock or multiple organ failure.

1.1.2 Virus transmission

Based on the data from World Health Organization, Ebola virus does not spread through the air, and no evidence has proved that the virus undergo the variation. Mode of transmission of Ebola virus is close direct contact with body fluids of patients, including the patient's blood, excrement, vomit which serve the strongest infection. Virus can also be found in saliva and tears.

1.1.3 Government research support

Federal government agencies by the US Food and Drug Administration (FDA) are directly under the charge of the US Department of Health and Human Services (DHHS) jurisdiction. Its main function is to supervise, certificate and manage the quality of US domestic production and imports of food, drugs, biological products, cosmetics, medical devices. It also plays a role in drug research funding support.

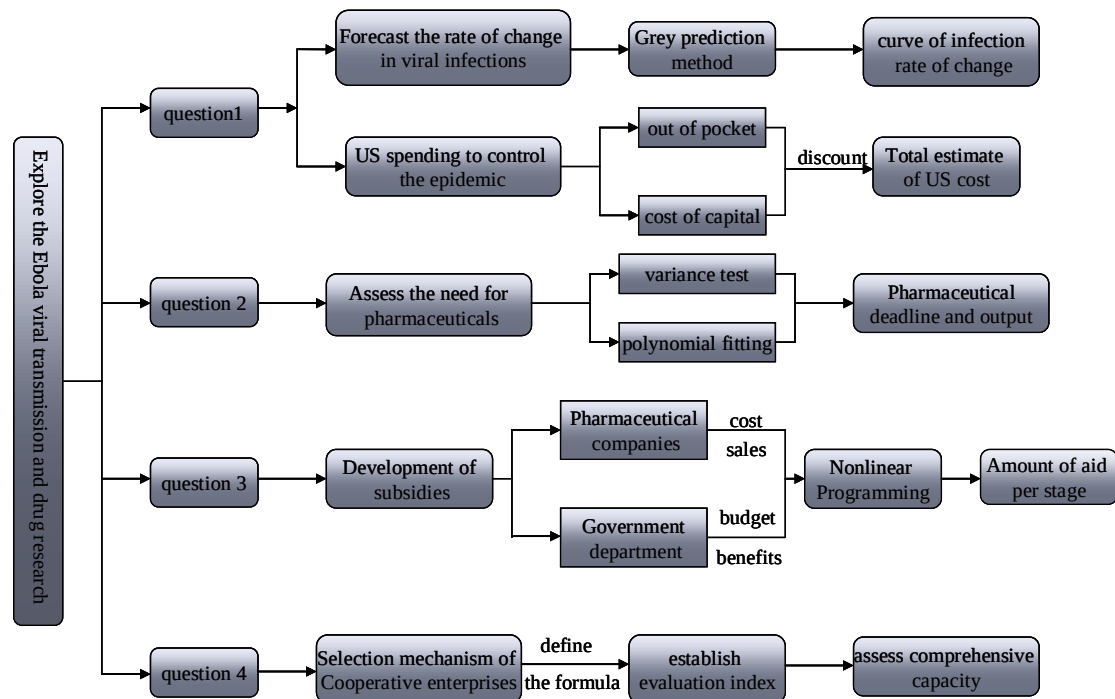
1.1.4 Research and development of new drugs

Overall, the development of new drugs is divided into two phases: research and development. These two stages are associated with each other one after another. The research phase includes four necessary parts: the identification of targets, establishment of model, discovery and optimization of lead compounds. To reduce the risk of drug development and costs of research, pharmaceutical companies will be combined in order to improve efficiency.

1.2 Problem restatement

We select Uganda as a typical region of Ebola virus infection in the African continent a mathematical model to predict the number of Ebola virus infection in the country from 2006 to 2025, and calculate the corresponding US investment required of the crisis. For the more, we analyzed during this time may be outbreaks of other infectious diseases, and to calculate the required response deadline for the infectious disease drug development and final volume of output. Finally, combined with 2013 R & D and the US budget, analyze government incentives for research and development of new drugs in order to maximize social benefits. In the end we provide more than one company co-developed programs to effectively reduce the cost of drug research and low risk.

1.3 Problem analysis



1.4 Symbols

Variables and their meanings

variable	Definition
C	Cost
s	The number of viral infection
k	The number of virus infection rate of change
T	Pharmaceuticals capacity
a	Scientific research capacity
b	Raw material supply capacity
c	Drug test experimental capability
d	Company cash flow ability

2 Analysis and Solution of Problem

2.1 Problem 1

2.1.1 Analysis of problem 1

We are required to select a country in Africa as a main research target on Ebola virus infections. According to the data from World Health Organization, we focus on Uganda. Based on the existing figure of Ebola infections in the year 2000, 2007, 2011, 2012, and 2013, we manage to obtain the coherent data from 2000 to 2014 through linear interpolation. Then we use gray prediction method to estimate the number of people at risk of infection from 2014 to 2025, and finally calculate the rate of change in the number of infections. For the United States' financial aid on this virus rage, we aim at the cost of the pharmaceutical investment. According to the opportunity cost of the different stages of pharmaceuticals available on the relevant literature, we assess the spending in the form of both cash and capital. Combined with the US discount rate

and growth of cost-consuming, we could figure out the average cost per drug in 2010.

2.1.2 Assumption of problem 1

a. The annual growth rate of out of pocket expenses and cost of capital remains stable after 1990.

b. The outbreak of the virus is intermittent, though the numbers of infections present a smooth trend during the week, and the patient can not heal.

c. We only target on the average expenditure on drug research and development.

2.1.3 Solution of Problem 1

Model I Ebola virus infection forecast model

2.1.3.1 Model preparation

a. Linear interpolation

If function $s(x)$ is continuous where $x \in [a, b]$, and at each interval $[x_i, x_{i+1}]$, function $s(x)$ is a polynomial, and $a = x_0 < x_1 < \dots < x_n = b$ are the certain pots, then regarding $s(x)$ as A spline function where x is $x_0, x_1, \dots, x_n$, if giving assumption $x = x_i$ where $y_i = f(x_i)$ ($i = 0, 1, \dots, n$), then:

$$s(x_i) = y_i, \quad i = 0, 1, \dots, n,$$

Regarding function $s(x)$ as a spline interpolation function.

b. Principle of grey forecasting methods

Firstly, assuming provide factors as:

$$x_i = \{x_i(1), x_i(2), \dots, x_i(n)\} \quad i = 1, 2, \dots, m$$

$$\Delta_i(k) = |x_0(k) - x_i(k)| \quad (k = 1, 2, \dots, n; 1 \leq i \leq m)$$

Mark standard data array x_0 's grey relational coefficient at k pots as

$$r_i = r(x_0, x_i) = \frac{1}{n} \sum_{k=1}^n (r(x_0(k), x_i(k))) \quad i = 1, 2, \dots, m$$

Secondly, build model: Assuming standard data array

$$x^{(0)} = (x^{(0)}(1), x^{(0)}(2), x^{(0)}(3), \dots, x^{(0)}(n))$$

Calculating the series ultima ratio:

$$\lambda(t) = \frac{x^{(0)}(t-1)}{x^{(0)}(t)} \in (e^{-\frac{2}{n+1}}, e^{\frac{2}{n+1}}) \quad (t = 2, 3, \dots, n)$$

If all ultima ratios $\lambda(k)$ lies in $\left(e^{-\frac{2}{n+1}}, e^{\frac{2}{n+1}}\right)$, then series $x^{(0)}$ could be regarded as

Model GM (1,1) to grey predicting. Otherwise we need to make switching to series $x^{(0)}$, making it lies in admissible region. That is to use a special count c as to making translation transform processing:

$$y^{(0)}(t) = x^{(0)}(t) + c \quad (t = 1, 2, \dots, n)$$

To make sure array $y^{(0)} = (y^{(0)}(1), y^{(0)}(2), \dots, y^{(0)}(n))$ passed by examination.

Thirdly, build model GM(1,1), to make accumulation of these array passed by ultima ratio examination, marking as:

$$y^{(1)} = (y^{(1)}(1), y^{(2)}(2), \dots, y^{(1)}(n)),$$

Use $y^{(1)}$ to make Mean Generating:

$$z^{(0)}(t) = 0.5y^{(1)}(t) + 0.5y^{(1)}(t-1) \quad t = 2, 3, \dots, n$$

Build array:

$$B = \begin{pmatrix} -z^{(0)}(2) & 1 \\ -z^{(0)}(3) & 1 \\ \dots & \dots \\ -z^{(0)}(n) & 1 \end{pmatrix}, \quad Y_N = \begin{pmatrix} y^{(0)}(2) \\ y^{(0)}(3) \\ \dots \\ y^{(0)}(n) \end{pmatrix}$$

Calculating:
$$\hat{u} = \begin{pmatrix} \hat{a} \\ \hat{b} \end{pmatrix} = (B^T B)^{-1} B^T Y_N$$

Obtaining the forecast:

$$\hat{y}^{(1)}(t+1) = (y^0(1) - \frac{b}{a})e^{-at} + \frac{b}{a} \quad (t = 1, 2, \dots, n-1)$$

Retrieving data:

$$\hat{y}^{(0)}(t+1) = \hat{y}^{(1)}(t+1) - \hat{y}^{(1)}(t) \quad (t = 1, 2, \dots, n-1)$$

2.1.3.2 Establishment and solution of the model

a. After using linear interpolation, the number of Ebola virus infections in Uganda is shown in Table:

Table 1 The number of Ebola infections each year							
year	2000	2001	2002	2003	2004	2005	2006
Ebola infection	425	386	347	307	268	228	188
year	2007	2008	2009	2010	2011	2012	2013
Ebola infection	149	128	75	38	1	24	7

It can be seen from the table that the series of data show a upward trend in general, from which we can learn that the epidemic situations have been relieved and controlled. Then we estimate the corresponding date from 2014 to 2025 through grey prediction method.

b. Assuming that the country's national health budget take a share $v\%$ of GDP, density of population as d (People/sq km), The national number for middle school students in the school as t' persons, The national educational level as t . We transform national number of middle school students in the school as the following:

according to practical circumstance, defining: $t = \frac{t - t'_{\min}}{t'_{\max} - t'_{\min}}$, regarding the solved data t as the educational level.

Build relationship between annual Ebola decreasing rate w and data v, d, t :

$$w = a_1v + a_2d + a_3t$$

Through the calculation of MATLAB:

$$s = 1273.3 - 13.56x_1 - 6.3134x_2 - 39.796x_3$$

This equation reflects the linear relationship among the relevant factors of Ebola virus infection. That means we could gain a better understanding of the constitution and development of the data.

After that, we adopt the gray prediction method to assess the number of Ebola virus infection in the following stage(2014~2025) through MATLAB programming.

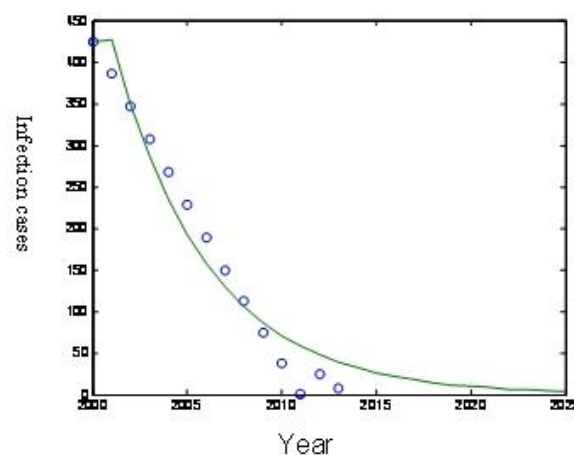


Fig 1 The estimation of the number of illness

Table 2 The estimation of the number of infection

Year	2000	2001	2002	2003	2004	2005	2006	2007	2008
Cases	425	426.38	349.37	286.27	234.57	192.2	157.49	129.04	105.74
Rate	0.0032	-0.1806	-0.1806	-0.1806	-0.1806	-0.1806	-0.1807	-0.1806	-0.1806
Year	2009	2010	2011	2012	2013	2014	2015	2016	2017
Cases	86.641	70.993	58.171	47.664	39.056	32.002	26.222	21.486	17.606
Rate	-0.1806	-0.1806	-0.1806	-0.1806	-0.1806	-0.1806	-0.1806	-0.1805	-0.1806
Year	2018	2019	2020	2021	2022	2023	2024	2025	
Cases	14.426	11.82	9.6855	7.9362	6.5028	5.3284	4.366	3.5775	
rate	-0.1806	-0.1805	-0.1801	-0.1806	-0.1806	-0.1806	-0.1806	-0.1806	

We can see from the table that the rate of change of Ebola virus infection tends to level off steadily. That means that with time passing by, Ebola virus disease can be effectively suppressed in the absence of other new drugs available.

C United States' Spending to deal with the crisis

According to the information of the product and research in the literature provided by the subject, we begin to estimate the approximate expenses for each type of drug annually from the initial phase 2000, to 2013.

Firstly, we divide the investment into two parts: out of pocket expenses and cost of capital which are the common ways of investment. Following this, we could

calculate the total cost of drug research and development which is correspondent with the investment.

Secondly, drug development process undergoes four stages: phase1 phase2 phase3 and animal test. To be specific, pre-clinical stage, clinical stage and testing stage constitute the whole process. In view of out of pocket funds allocated in each stage of the drug development and rate of approval for each stage, the total cost of drug development until the advent of the new drug can be determined by multiplying the cost of each phase and the success rate respectively. (unit: millions).The table is shown as follows:

Table 3 relevant information of OOP cost

	Mean cost	Probability of entering phase(%)	Expected cost
Phase I	15.2	100%	15.2
Phase II	23.5	71%	16.7
Phase III	86.3	31.4%	27.1
Long-term animal	5.2	31.4%	1.6
Total			60.6

Therefore, the average investment in out of pocket stage to develop a new drug is \$ 60.6million.

In the respect of capital investment, average cost of capital in each stage is stable. We could calculate the figure of input via the calculation method we have established above(the average cost of capital × throughput rate per stage). The result here is \$100.4 billion.

Therefore, the total cost price which is the summation of out of pocket expenses and cost of capital is \$161million in 2000 dollars.

Based on the growth rate of out of pocket cost and cost of capital per year(7.6%and 7.4% respectively), we explore the calculating formula to estimate the expenditure of investment with the benchmark of data in 2000.

$$C = C_{OOP} + C_{COC} = 60.6(1 + 7.6\%)^{T-2000} + 100.4(1 + 7.4\%)^{T-2000}$$

However, the new drug is not necessarily put into the market successfully even though it has completed long-term animal experiments. In the mean time, from the results above we could know that the actual cost is mainly the final cost put into the drug market. That means the cost in mid-experiment on drug research which occupies much of the overall cost is not included into the recorded expense. Such situation increases the error with no doubt.

The probability of a new drug to put into the market is recorded as 21.5% .So the average cost of each drug to deal with each disease is evaluated by the equation

$$C' = \frac{C}{21.5\%}$$

The cost of each type of drug is shown in the following table on the basis of 2000 dollars.

Table 4 Estimation of cost for each year

Year	2000	2001	2002	2003	2004	2005	2006	2007
Cost	748.84	804.81	864.98	929.64	999.13	1073.83	1154.10	1240.38

Year	2008	2009	2010	2011	2012	2013	2014	
Cost	1333.11	1432.77	1539.89	1655.01	1778.75	1911.73	2054.66	

Considering that the value of the dollar is changing constantly, we calculate the expenditure of drug research from 2000 to 2003 with the addition of the discount rate considerations (the data of US dollar discount rate is only available until 2013). The exact cost from 2000 to 2013 is stated as follows:

Table 5 discount cost in drug production for each year

Year	2000	2001	2002	2003	2004	2005	2006
Cost	748.84	850.93	917.38	987.15	1062.39	1144.66	1234.77
Year	2007	2008	2009	2010	2011	2012	2013
Cost	1332.16	1434.11	1541.87	1657.98	1785.99	1925.34	2076.48

It can be seen from the estimation, the US governmental spending used to deal with the epidemic crisis shows an upward trend year by year, and this figure is up to \$2076.48million in 2013.

2.2 Problem 2

2.2.1 Analysis of problem 2

The problem requires us to determine whether new drugs are needed during 2006 to 2025 in Uganda. We choose more significant types of diseases in the region: tuberculosis, AIDS and Malaria. If there is a significant difference in variance test, the need for new drugs is obvious. After the determination of the need of new drugs, we extract the outliers' number of infections per year for each disease. The total output of pharmaceutical drugs were estimated based on the exact number of infections during the certain period.

2.2.2 Assumption of problem 2

- We don't consider cross-impact caused by virus mutates or other virus;
- Assuming unit output as the average dosage of curing a patient.

2.2.3 Solution of Problem 2

Model II Necessity model of drug development

2.2.3.1 Model preparation

The specific steps of fitting curve are below:

With the given data $(x_i, y_i) (i = 0, 1, \dots, m)$, and formula Φ , we decided to figure out

$$p(x) \in \Phi \text{ to minimize error } r_i = p(x_i) - y_i (i = 0, 1, \dots, m)$$

$$\sum_{i=0}^m r_i^2 [p(x_i) - y_i]^2 = \min$$

From geometry meaning is seeking for the curve $y = p(x)$ that has the shortest distance with given point $(x_i, y_i) (i = 0, 1, \dots, m)$, Function $p(x)$ is known as the fitting function or the least squares solution. The method is called curve fitting least square method.

And the method of polynomial fitting can be summarized as the following steps:

- ① Drawing rough graphics by the known data rough graphics (scatter plot), determining the number of fitting polynomial;

② Calculating $\sum_{i=0}^m x_i^j (j = 0, 1, \dots, 2n)$ and $\sum_{i=0}^m x_i^j y_i (j = 0, 1, \dots, 2n)$

③ Giving a formal system of equations, calculating $a_0, a_1, \dots, a_n$;

④ Providing polynomial fitting $p_n(x) = \sum_{k=0}^n a_k x^k$

b. ANOVA test

The first step is a clear observation and control variables.

The second step is to analyze the variance of the observed variables. Variance analysis believes that: changes in the observed value of the variable will be affected by the control variables and random variables. Accordingly, the ANOVA analysis observed variables and the total sum of squares decomposition between groups from within the sum of squares and groups sum of squares of two parts, with the mathematical form expressed as: $SST = SSA + SSE$.

The third step is to observe the total sum of squares proportional and comparing the observed variables to infer whether the observed variables brings significant impact.

2.2.3.2 Establishing and solving of the model

a. To solve this problem, we can study the fluctuations of other infections based on the number of Ebola infections. We select more significant infectious diseases in Uganda: tuberculosis, AIDS, and Malaria. Tuberculosis and AIDS data are more complete range, while Malaria has been missing. So we use polynomial approximation to estimate infections in other years.

On the basis of infection cases of Malaria from 2000 to 2008 in Uganda, we estimate the number of patients from 2009 to 2012 by using polynomial fitting. Specific data of patients is shown as following Figure2:

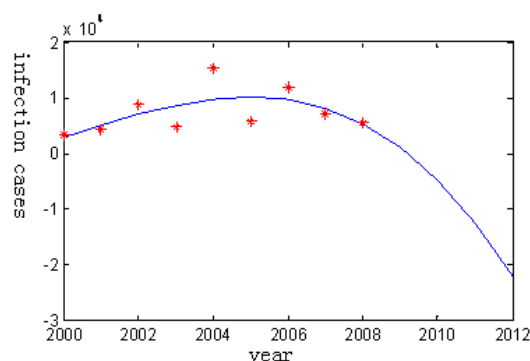


Fig 2 number of infection cases during the period

Table 6 number of infection cases during the period

Year	2000	2001	2002	2003	2004	2005	2006
Cases	3070.4	5164.4	7107.5	8716.6	9808.7	10201	9709.6
Year	2007	2008	2009	2010	2011	2012	
Cases	8152.3	5345.9	1107.3	-4746.6	-12399	-22032	

Thus, we have the number of (Ebola, dengue fever, tuberculosis and HIV) infections each year in Uganda, the disease can be seen by the number of patients in a

controlled state of illness in 2010 the number reached a negative net worth, said to be cured. Specific data is shown as following Figure3:

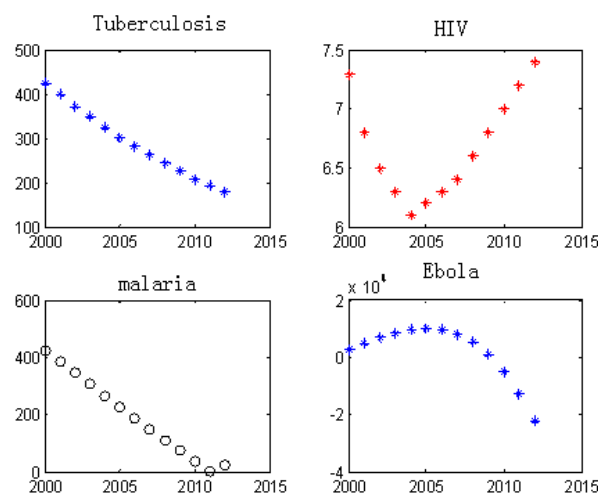


Fig 3 number of infections for different diseases

b. We analyzed number of patients with Malaria, tuberculosis and AIDS as a simple weighted. In this case, in order to determine whether requires the development of new drugs, we estimated Ebola infection data with previous data by variance test. If there are significant differences, the disease has not been controlled as Ebola control, and has the trend of going worse. So there is need for developing new drugs. If there is no significant difference, there is no need to develop new drugs.

Specific analysis steps:

①Propose original hypothesis:

H_0 : there is no obvious difference between overall infection and Ebola

H_1 : there is obvious difference between overall infection and Ebola

②Select test statistics, which is F test,

③Calculation of test statistics observed value and probability values,

④set significant level: $\alpha = 0.05$

We have conclusion below:

Source	SS	df	MS	F	Prob>F
Columns	2.84974e+013	1	2.84974e+013	529.97	0
Error	1.29053e+012	24	5.37722e+010		
Total	2.9788e+013	25			

Fig 4 ANOVA information

From the chart, we can conclude that, $F = 529.97$ which is a large number, and $p = 0 < 0.05$ which passes the significance test that significant differences exist between two data, which means there is need to develop a new drug.

c. For the latest period for drug development and possible production, we observe annual infections to identify abnormal data of fluctuation as a basis for the development. Then the drug development cycle could control the disease before the

outbreak of the next period, enabling it to be put into the market.

The overall number of people suffering from infectious diseases make up the fluctuation map by MATLAB, as the Figure below:

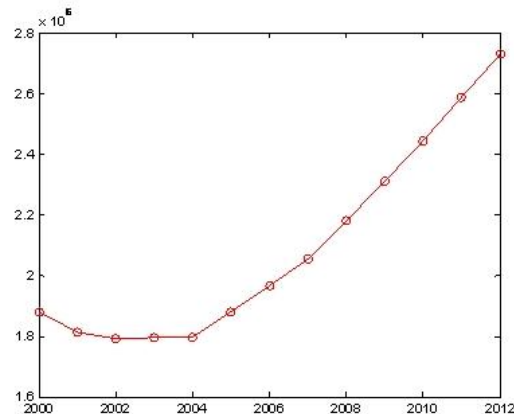


Fig 5 number of sickness during the period

From the figure we can see that the infection cases before 2004 was smoothly, but 2004 experienced a sudden increase, which indicate that at this time there is a new outbreak of the epidemic and have the need of developing new drugs for the start time. After 2004, the number of patients continues to grow, this time by comparing rates of change in different years, the number of patients to determine further deterioration in what year, we set the number of infection rate of change:

$$k(i) = \frac{s(i) - s(i-1)}{s(i)}$$

Sequence $s(i)$ for $(i = 1, 2, \dots, 12)$ are every year's infections, we conclude the curve graph below::

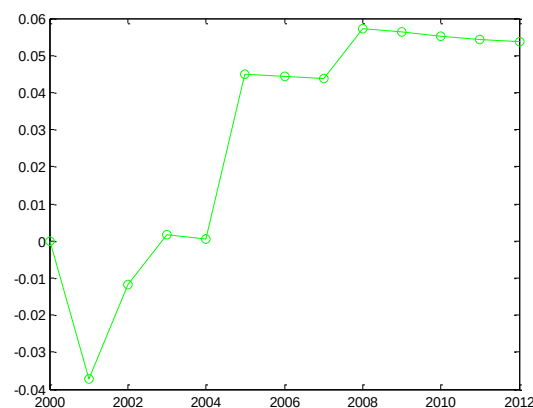


Fig 6 change of rate in number of infections

As we can see from the figure, the infection rate of change reached the maximum rate in 2008 when new drugs must invest to control the disease. The maximum period of the new drugs development is 4 years.

Based on the demand to produce the drug before 2008, we could estimate the approximate output of drugs as 2.2 million which is equal to the figure of infections till 2008.

2.3 Problem 3

2.3.1 Analysis of problem 3

Problem requires us to devise a scheme to provide financial support for pharmaceutical companies on the process of drug development. This program is aimed at reducing drug prices and compressing the drug sales so that those poorer countries in Africa could afford the expense and drugs can not be less effective due to the wide spread of usage. We would divide the subsidies to pharmaceutical companies into two stages: drug research and development stage and sales stage. Government subsidies are derived from the government's own budget and public health benefits. In order to facilitate the calculation, we will capitalize the public health benefits, thus obtaining the best assistance programs.

2.3.2 Problem assumption 3

- Supply-demand relationship does not affect the change of drug price: decline of price will not lead to higher demand for drugs;
- The governmental health budget are completely put into the drug research and development;
- Output is equal to the sales, there is no stock;
- Companies are reluctant to develop tropical medicine only because of the uncertainty of cost-recovering.

2.3.3 Solving of problem 3

Model III Reward program device model

2.3.3.1 preparing model

- Theory: nonlinear programming method

The general form of mathematical programming model:

$$\begin{cases} \min f(x) \\ \text{s.t. } g_i(x) \leq 0, \quad i = 1, \dots, p \\ \quad h_j(x) = 0, \quad j = 1, \dots, q \end{cases}$$

$x = (x_1, x_2, \dots, x_n)^T$, $f(x), g_i(x), h_j(x)$ are real functions based on x .

The introduction of vector function symbols:

$$h(x) = (h_1(x), \dots, h_q(x))^T \quad g(x) = (g_1(x), \dots, g_p(x))^T$$

$$X = \left\{ x \in R^n \begin{cases} g_i(x) \leq 0, i = 1, \dots, p \\ h_j(x) = 0, j = 1, \dots, q \end{cases} \right\}$$

$$\begin{cases} \min f(x) \\ \text{s.t. } g_i(x) \leq 0, \quad i = 1, \dots, p \\ \quad h_j(x) = 0, \quad j = 1, \dots, q \end{cases} \Rightarrow \begin{cases} \min f(x) \\ \text{s.t. } g(x) \leq 0 \\ \quad h(x) = 0 \end{cases} \Rightarrow \min_{x \in X} f(x)$$

When there is at least on function of function $f(x), g_i(x), h_j(x)$ is nonlinear model, fitting the nonlinear programming (NLP)

Claims:

$$X = \left\{ x \in R^n \begin{cases} g_i(x) \leq 0, i = 1, \dots, p \\ h_j(x) = 0, j = 1, \dots, q \end{cases} \right\}$$

As MP constraint set or the feasible region of the problem. If x is within the space of X , Said x the feasible boot or feasible points in MP.

As for inline program (MP), if $x^* \in X$, at the same time

$$f(x^*) < f(x), \forall x \in X$$

Then regard x^* as the overall optimal boot or the overall minimum point of [MP].

b. The data processing

① the relevant data of pharmaceutical company

The average cost to develop a new drug takes \$2076million according to the information from the relevant literature and the survey from related pharmaceutical department.

② United States budget and public health department outlay

The U.S. government budget is: \$1040million ; Public health care spending of R&D is \$383million .

③ Unknown data to determine

Due to limited drug sales, drug price and sales are unknown. The figure of aid spending from government and public health department is undecided.

2.3.3.2 establish and solve of the model

a. In view of the two stages of the program , we should make sure that each stage is targeted on the different aspects. The pharmaceutical company are reluctant to take the risk of depressing the price of drug to meet the demand of African market. So the only initiative to carry out the drug research is to be rewarded financially. In addition, when drug producing finishes, pharmaceutical companies are faced to drug sales, thus weakening the patient's body resistance. Therefore the aid for the second phase is the compensation for inhibition of sales.

After that, we need to set required constraints for each variable:

① In order to simplify the calculation, we only device the scheme in 2013. The drug production minimum should satisfy the demand of starting point of the outbreak in 2004, namely \$1.9million (each person need a pair of medicine). In the meantime, in order to meet the requirements of production limit, set up the maximum as the figure of sickness in 2008, namely \$2.2million.

② The aid for the first phase of the pharmaceutical companies can be considered as a subsidy to the company's cost of drugs. We set the range of the amount of compensation from 20% and 40% of the cost.

③ To calculate the net deficiency of drug research, we should take the company's sales into consideration. We use the expression: $C-S$ to represent the loss. After enjoying the government assistance, the company may still suffer from the loss. So we set up the total compensation scope from 50% of $C-S$ to 120% of $C-S$.

④ In order to make full use of the benefits of public resources and the budget in medical treatment, we also set a lowest limit on the investment. i.e.: the amount of

fund must be up to 20% of the total.

b. nonlinear programming model is set up:

Set x_1, x_2 respectively for the first and second stage pharmaceutical companies' subsidies. Set x_3, x_4 respectively for the cost of the government budget and health welfare funds consumed by the department of public. Divided into two parts, set x_{31}, x_{32} respectively for the first stage consuming budget and second stage of that. then x_{41} as first stage of public health welfare funds and x_{42} as second stage's. Set x_5, x_6 respectively as drug price and yield.

Firstly, the less Pharmaceutical companies could losses in the process of producing, the more government compensation can be, The higher the satisfaction is, we assuming:

$$d_1 = \begin{cases} \frac{x_1 + x_2}{2.076 \times 10^9}, & x_1 + x_2 < 2.076 \times 10^9 \\ 1, & x_1 + x_2 \geq 2.076 \times 10^9 \end{cases}$$

As for government departments and public health departments, the less they spend, the more satisfied they are, according to the setted budget limit, we can set satisfied index as follows:

$$d_2 = \begin{cases} 1, & x_3 \leq 20\% \times 1.04 \times 10^9 \text{ \& } x_4 \leq 20\% \times 1.04 \times 10^9 \\ \frac{20\% \times 1.04 \times 10^9}{x_3} + \frac{20\% \times 3.83 \times 10^8}{x_4}, & x_3 > 20\% \times 1.04 \times 10^9 \text{ or } x_4 > 20\% \times 1.04 \times 10^9 \end{cases}$$

At the same time, due to the limitation on the drug production, we add another objective index d_3 , it defines as follows:

$$d_3 = \begin{cases} 1, & x \leq 1.9 \times 10^6 \\ \frac{1.9 \times 10^6}{x_6}, & x > 1.9 \times 10^6 \end{cases}$$

Here is nonlinear programming model:

$$\begin{aligned} \max \quad & d = \frac{1}{3} \times d_1 + \frac{1}{3} \times d_2 + \frac{1}{3} \times d_3 \\ \left\{ \begin{array}{l} d_1 = \frac{x_1 + x_2}{2.076 \times 10^9}; \\ d_2 = \frac{20\% \times 1.04 \times 10^9}{x_3} + \frac{20\% \times 3.83 \times 10^8}{x_4}; \\ x_{31} + x_{32} - x_3 = 0; \\ x_{41} + x_{42} - x_4 = 0; \\ x_{31} + x_{41} - x_1 = 0; \\ x_{32} + x_{42} - x_2 = 0; \\ x_5 \times x_1 - x_6 \times x_2 = 0; \\ 0.5 \times (2.076 \times 10^9 - x_5 \times x_6) \leq x_1 + x_2 \leq 1.2 \times (2.076 \times 10^9 - x_5 \times x_6) \\ 20\% \times 1.04 \times 10^9 \leq x_3 \leq 1.04 \times 10^9; \\ 20\% \times 3.83 \times 10^8 \leq x_4 \leq 3.83 \times 10^8; \\ 20\% \times 2.076 \times 10^9 \leq x_1 \leq 40\% \times 2.076 \times 10^9; \\ 1.9 \times 10^6 \leq x_6 \leq 2.2 \times 10^6; \\ d_3 = \frac{1.9 \times 10^6}{x_6} \end{array} \right. \end{aligned}$$

Though the LINGO software, we calculate that the amount of subsidies in the first stage is \$415.2 million, including government departments at \$208 million, public health departments at \$207.3432 million. The second stage of the reward is \$143228.3 million, including the government departments at \$2006194 million, public health departments at \$143228.1 million. In the meantime, the drug price per unit is \$655.43 million and the output is 1.9 million.

It can be seen from the program that the amount of total spending from US budget and public health service is \$210 million and \$207.34 million. From this conclusion, the government aid account for only 20% of the total budget, which is close to the minimum. By contrast, public health officials took a relatively larger amount, close to its limit. From a practical sense, this part of funding is for social welfare, health care and other non-monetary properties of subsidies in pharmaceutical enterprise itself, such as free trial of new drugs, supply of free annual check-up services, etc.

2.4 Question 4

2.4.1 The analysis of question 4

Question four require the government to design a scheme to select company cooperate drugs to reduce the cost and risk. In order to effectively play to the advantages of different companies, give play to their strengths in the pharmaceutical industry, based on the company's related data, we define four indicators to assess their drug research and development ability. , according to the various parameter values for each company, selecting the most advantage of each company to mix and match. Finally, as for the consideration of sustainability, we take the effect caused by increased number of companies into account.

2.4.2 The foundation of model 4:

Model IV Cooperative companies selecting model

Considering reducing costs and risks effectively by joint pharmaceutical companies, we need to select some of the lowest cost and risk in the process of the

pharmaceutical part to be combined, the source of these parts is the optimal joint companies.

We define Scientific research ability as a , the ability to produce pharmaceutical materials as b , the ability for pharmaceutical inspection as c , Drug late tracking and feedback ability as d .

Assume that there are n companies involved in pharmaceutical, Establish comprehensive pharmaceutical capacity formula as follow :

$$T = \lambda_1 \sum_{i=1}^n a_i + \lambda_2 \sum_{i=1}^n b_i + \lambda_3 \sum_{i=1}^n c_i + \lambda_4 \sum_{i=1}^n d_i \quad \sum_{i=1}^4 \lambda_i = 1$$

After consulting literature material, gives the calculation formula of the following four indexes(a 、 b 、 c 、 d) and study how the four indicators reducing the effects of drug cost and the risk.

Scientific research ability of the quantitative measuring model:

For any pharmaceutical company number i , define x_1 as the quantities of research staff, the company's total staff number as X , the average speed of working on a new drug as v_1 , the average number of patent applications every year as t , The equation is as follows:

$$a'_i = \frac{x_1 v_1 t}{X}$$

Define y_1 as the company's nearly three years' average annual production quantity of drug, Y as the total quantity of the drug at the year, v_2 as the speed to averagely produce one ton of the drug.

$$\begin{cases} b'_i = \frac{y_1 v_2}{Y} \times t \\ t = \frac{y_1 - y_{\min}}{y_{\max} - y_{\min}} \end{cases}$$

Define y_s as the actually sales quantity on the year, y as the quantity of the drug produced of the year, x_2 as the quantity of inspection personnel:

$$\begin{cases} c'_i = \frac{y_s}{y} + N \frac{x_2}{x_1} \\ N = 2, 3, 4, \dots \end{cases}$$

Define y , c respectively the company's turnover per year on average of the nearly three years and average costs of the nearly three years, r as turnover rate for the company, R as average turnover rate of the all players:

$$d'_i = \frac{Z}{2M} (g - c) + \frac{r}{2R}$$

As for Comprehensive pharmaceutical ability :

$$, \quad T = \lambda_1 \sum_{i=1}^n a_i + \lambda_2 \sum_{i=1}^n b_i + \lambda_3 \sum_{i=1}^n c_i + \lambda_4 \sum_{i=1}^n d_i \quad \sum_{i=1}^4 \lambda_i = 1$$

transform the indictors a'_i, b'_i, c'_i, d'_i as follows:

$$\left\{ \begin{array}{l} a_i = \frac{a'_i}{\sum_{i=1}^n a'_i} \\ b_i = \frac{b'_i}{\sum_{i=1}^n b'_i} \\ c_i = \frac{c'_i}{\sum_{i=1}^n c'_i} \\ d_i = \frac{d'_i}{\sum_{i=1}^n d'_i} \end{array} \right. \circ$$

Bring the data in this formula: $T = \lambda_1 \sum_{i=1}^n a_i + \lambda_2 \sum_{i=1}^n b_i + \lambda_3 \sum_{i=1}^n c_i + \lambda_4 \sum_{i=1}^n d_i$,

Respectively calculated comprehensive abilities of different combination of integrated pharmaceutical companies, if $i \rightarrow \infty$, $T \rightarrow \infty$, But at this time and the cost of labor force will grow indefinitely.

To establish a method to limit the companies' number growing: within limits,

$i = 1, 2, \dots, m \dots k$ ($k \geq m$), with the number of i growth, the number of T is increasing at the same time.

To calculate the comprehensive ability of any companies:

$$T = \lambda_1 a_i + \lambda_2 b_i + \lambda_3 c_i + \lambda_4 d_i$$

Then sort these companies according to the comprehensive ability in order:

$T_m \geq T_k \geq T_l \geq \dots T_{\min}$, to sum from small to large: $\sum_{i=1}^n T_i, \sum_{i=1}^{n+1} T_i, \dots \sum_{i=1}^{n+k} T_i$, construct growing sequence as the add of company's quantity: $y_1, y_2, y_3 \dots y_{n+k}$. Calculating the growth of speed ρ , setting a threshold value η , stop to add the amount of companies when $\rho \leq \eta$:

$$\left\{ \begin{array}{l} \rho = \frac{y_{i+1} - y_i}{y_i} \\ \rho \leq \eta \end{array} \right.$$

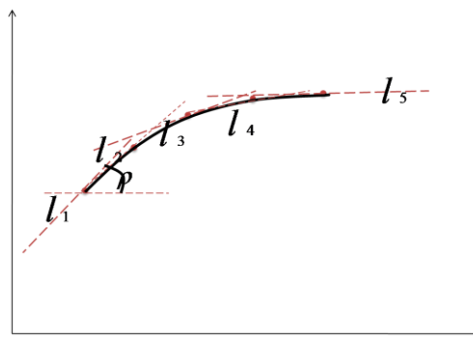


Fig.7 ρ Variation diagram

So, when $\rho' < \eta'$, choosing a combinations of $y_1, y_2, y_3 \cdots y_i$ to comprehensively manufacturing medicine, it also need to join other constraints According to the actual situation, Here is no longer explaining.

3. Error Analysis

3.1 Error Analysis of question 1

Gray prediction is the method of using past data to predict the future. In question 1, we selected Ugandan infections from 2000 to 2013 which is not completely. And we use the linear interpolation to calculate the lack of data from 2000 to 2013, and use the gray prediction method to forecast at the same time to estimate Ugandan infection from 2000 to 2025. That is where the error comes.

3.2 Error Analysis of question 2

To determine whether there are needs for new drugs, we compare infections with Ebola infections to observe differences. But in fact, the infection is not the only standard to measure, there may be other factors which can promote the drug research and development, such as national policy, scientific research needs, etc. As a result of these indicators is not easy to quantify. We will not be considered in our paper. So there will be some error.

3.3 Error Analysis of question 3

We use data in 2003 to provide government a subsidy project to pharmaceutical companies. And to simplify the problem we set the upper limit and lower limit value on the basis of experience. But actually, the various data fluctuation will change from time to time, and the critical value is not the stable. So resulting may have some errors.

3.4 Error Analysis of question 4

There may be another company B has one prominent ability then A, and in the combination of A few only without B have this ability, then choose B than use A higher overall efficiency. But when $T_A \geq T_B$, there will be some errors which need to be corrected when the amount of companies is small. But when $\Delta T / T \rightarrow 0$, the error can be ignored.

4 Sensitivity analysis

In question 3, we set some parameters to determine the subsidy scheme. But in the actual cases, these parameters will exist. Therefore, we select subsidies account for a net loss of the percentage k , the lower limit of the government budget z and government subsidies percentage threshold adjusted respectively, then observe the changes of plan related numerical.

We choose k as 0.3, 0.4, 0.5, 0.6 and 0.7 respectively to observe the changes of drug price and the second phase compensation

Table 7 table 1 of sensitivity analysis

k	0.3	0.4	0.5	0.6	0.7
<i>price per unit</i>	364.07	546.16	655.43	728.28	780.32
<i>reward in phase two</i>	79559.08	119350	143228.3	159148.7	170521.1

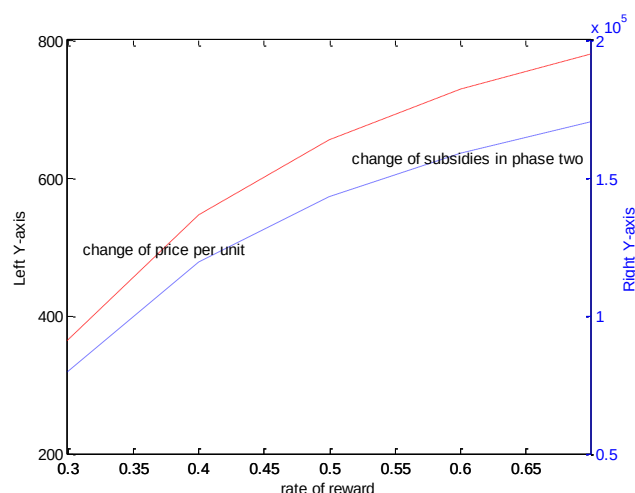


Fig 8 graph 1 of sensitivity analysis

We can conclude from the figure that: When the lower limit of is gradually increasing, the drug price and the second stage subsidies are gradually increasing, and presents the trend of slow, which shows that the elasticity of price decrease after reaching threshold value.

For another parameter z as 0.1, 0.2, 0.3, 0.4, 0.5 and 0.6, to observe the change of the expenditure of the government. Specific data are shown below:

Table 8 table 2 of sensitivity analysis

z	0.1	0.2	0.3	0.4	0.5	0.6
number	2.08×10^9	2.08×10^9	3.39×10^9	9.09×10^9	7.54×10^9	7.54×10^9

And we conclude the curve at the same time:

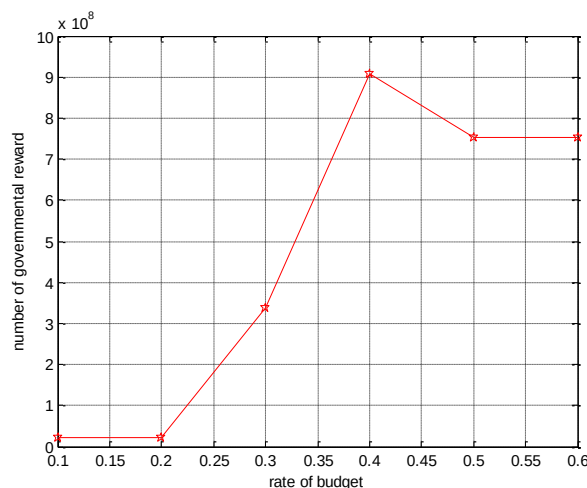


Fig 9 graph 2 of sensitivity analysis

From the curve we can see that: this is different from the previous one. When parameter is increase from 20% gradually, the government's budget is increasing at a increasing rate. But the after that the government budget is decreasing and maintain a stable situation after 50% of budget.

5 Evaluation of models

5.1 Advantages of the models

5.1.1 The nonlinear model aimed at government compensation problem not only reduce the costs of pharmaceuticals companies, but also saves lots of government budget. in addition it analyzes the interactions among them.

5.1.2. The accuracy of the data has been fundamental to ensure cause the cost of the study are the present value of the discount.

5.1.3. The number of predicted infections through image has a strongly intuitive recognition.

5.1.4. Comprehensive use of matlab, Eviews software, enhancing the reliability of the mathematical model .

5.2 Disadvantages of the model:

5.2.1. Ignore the change rate of drug research each year ,there is a certain deviation.

5.2.2. In this paper, the scheme we give is only based on the 2013 budget, the objective effect of guidance is not obvious.

6 Reference

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